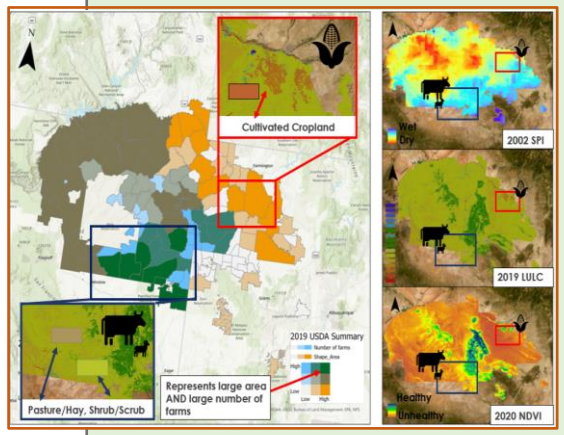
The Story Water Holds: Utilizing Earth Observations For Resource Management and Story Telling

M.René Castillo[1], Michael Durand[2], Wendy Panero[3]

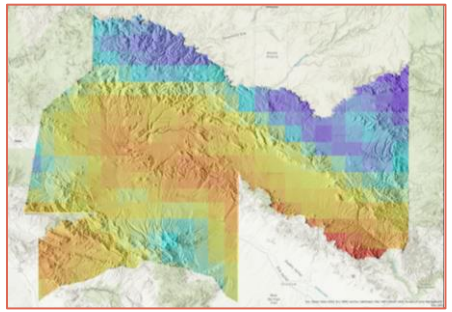
Modern remote sensing techniques improve drought monitoring, identification, and story telling.



Comparison of multiples remote sensing indices across the Navajo Nation to understand drought signatures through remote sensing, RC 2023.

Remote Sensing

- Radiation is reflected, absorbed, and emitted by the Earth's atmosphere or surface.
- Satellites carry sensors measuring electromagnetic radiation reflected or emitted from terrestrial and atmospheric sources.
- Scientists measure height, temperature, moisture content (and more) for various features of Earth's systems.
- This information can be visualized to inform about systems and their impacts.



How can you start?

Study Area

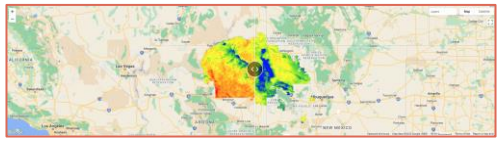
Remote sensing image showing the annual precipitation values compared to the terrain, RC 2023.

App Development

Images from remote sensing web application developed for the Navajo Nation, RC 2023.

Interest: streamlining visualization

Spectral Analysis: chart production



Product: comparison map, faster data development

In Honor of Mary Milwee: Storyteller, Mother, Grandmother

Voices of the community

QR CODE to Documentary here

Story Telling

Observations have qualitative depth alongside numerical data. Indigenous scientists use storytelling to weave a holistic understanding beyond data points.

Data visualization enhances narratives by making complex information accessible.

Indigenous communities emphasize collaborative storytelling, urging scientists to integrate traditional knowledge, and ensure cultural accuracy. Scientific stories can be robust and culturally inclusive.

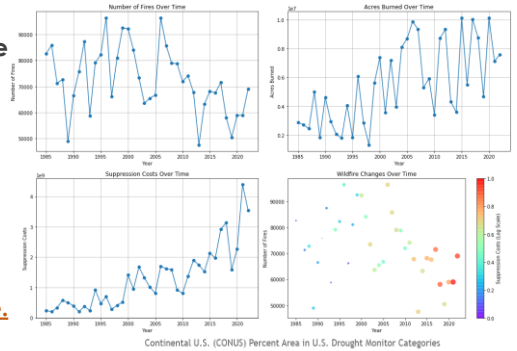
Who is being impacted?

What data is accessible?

What is important?

What is The Story?

- Indigenous communities are the most vulnerable!
- Drought is a problem of moisture imbalance at the land surface.
- Drought influences multiple hazards.



Wildfire comparison, the bubble chart shows all variables compared against each other with the acres burned determining bubble size, color of the bubbles represents the normalized cost of suppression (more costly towards 1.0), y-axis the number of fires against the x-axis time, RC 2023.

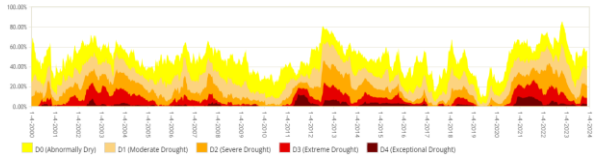


Image Credit: U.S. Drought Monitor

The Ohio State University occupies is the ancestral and contemporary territory of the Shawnee, Potawatomi, Delaware, Miami, Peoria, Seneca, Wyandotte, Ojibwe, and Cherokee peoples. I would like to acknowledge these tribal nations and recognize the historical contexts that have and continue to affect the Indigenous peoples of this land.

I would like to acknowledge that the Region of Interest for this project is the traditional homelands of the original peoples of New Mexico, Arizona, and Utah – Pueblo, Navajo, and Apache . I honor the land and those who steward the land throughout the generations and acknowledge their continued struggles..

I would like to acknowledge that this work was funded by the Friends of Orton Hall grant from The Ohio State University School of Earth Sciences and The Institute for Tribal Environmental Professionals at Northern Arizona University.

EGU 2023 AG2023 RC 2023